



An interpretable Dictionary Learning approach for Diagnostic Hypotheses from 3D Chest

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Abstract

Automated Diagnostic Hypotheses from 3D chest Computed Tomography requires clinically interpretable methods to support radiological decision-making. Current state-of-the-art (SotA) deep learning models often rely on post-hoc explainability techniques that provide approximate localizations and lack direct traceability to the models decisions. Furthermore, SotA models rarely support simultaneous multi-class classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning framework in which detected abnormalities can be back-projected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset with expertly annotated abnormality localizations. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

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Acronyms

SotA	-	State-of-the-Art
CT	-	Computed Tomography
AI	-	Artificial Intelligence
VLFM	-	Vision Language Foundation Model
KSVD	-	K-Means Singular Value Decomposition
LC-KSVD	-	Label Consistent K-SVD
HU	-	Hounsfield Unit
DFDL	-	Discriminative Feature oriented Dictionary Learning

Chapter 1

Introduction

Developing a unified architectural pipeline for multi-class abnormality classification and grounded localization in 3D Computed Tomography (CT) addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is held accountable and its reasoning aligns with clinical diagnosis.

1.1 Background

The transition from 2D projectional radiography to volumetric CT represents a paradigm shift in medical image analysis.

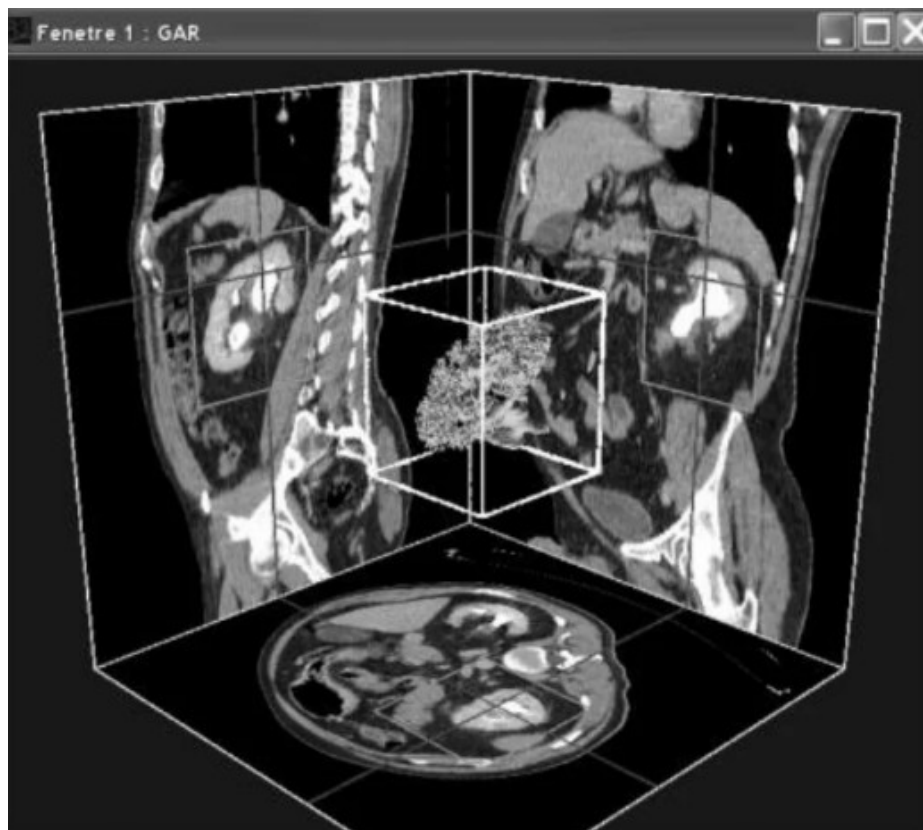


Figure 1.1: Transition from 2D projectional radiography to volumetric CT.

For Lung disease diagnosis, Chest X-rays (CXR) are commonly used as the first imaging test due to its low cost and low radiation exposure. However, studies show that their capability to detect abnormalities is limited when compared to 3D CT. [1].

Furthermore, CXR has historically benefited from the availability of massive, radiologist-annotated datasets, while the analysis of chest CT volumes has been severely constrained by a lack of such data. However, the recent introduction of large-scale, text-paired datasets has facilitated the development of state-of-the-art

(SotA) vision-language foundation models tailored specifically for volumetric imaging [2–4].

Beyond deep learning approaches, Dictionary Learning algorithms, particularly, Label Consistent KSVD (LC-KSVD) [5], Label Embedded Dictionary Learning (LEDL) and Fischer Discriminant Dictionary Learning (FDDL) offer viable approaches for interpretable CT analysis. All three techniques incorporate label information during the dictionary training process, and ensures that the sparse representation of the data is highly discriminative, making them well-suited for multi-abnormality tasks.

1.2 Problem Statement

The clinical utility of Deep Learning-based approaches is hindered by a lack of interpretability. While tools like Grad-CAM provide post-hoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model’s decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance.

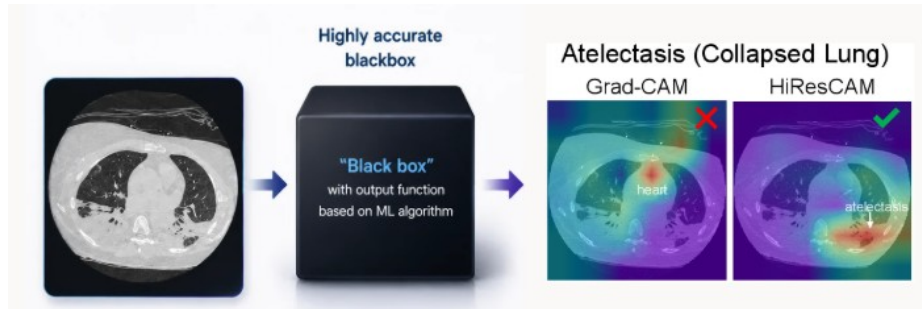


Figure 1.2: HiRes-CAM example illustrating higher-resolution localization compared with Grad-CAM.

1.3 Objectives and Scope

The primary goal of this project is to address the limitations of existing Deep Learning based approaches for Lung abnormality classification and localization by developing an inherently interpretable dictionary learning framework. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning approach to learn normal and pathological dictionary atoms using a publicly available annotated 3D chest CT dataset.
- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.
- Benchmark model performance against state-of-the-art classification and segmentation models on the same dataset.
- Evaluate abnormality localization by the model on a real-world dataset with expert radiologist review.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

- This study is limited to 3 thoracic abnormalities: Pulmonary Nodules, Ground-Glass Opacities, and Consolidations, based on expert radiologist recommendation and dataset availability.
- This study will utilize a single publicly available dataset for training and evaluation, which may limit generalizability to other populations and scanner types.
- The proposed approach will be limited only to linear methods, and will not explore non-linear or hybrid models that provide better performance for less explainability.

1.4 Literature Review

The landscape of 3D CT analysis is currently divided into three primary architectural paradigms:

- Classification-based models that prioritize diagnostic labels
- Segmentation-based models that prioritize boundary precision
- Unified architectures designed for grounded localization

SotA models are predominantly specialized in either classification or segmentation. However, recent systems, particularly in 2025 and 2026, are moving toward multi-task learning or foundation models that can perform both diagnosis and voxel-level segmentation simultaneously.

Table 1 summarizes the key models in each category, their primary task orientation, and their performance on standard benchmarks.

Table 1.1: Summary of major 3D CT analysis paradigms and representative models

Paradigm	Representative Models	Primary Task	Typical Strengths / Notes
Classification-based	CT-CLIP; Merlin; DINOv3	Image-level diagnosis; retrieval; linear probing	Strong image-level diagnosis; good zero-shot and transfer performance; efficient inference; limited voxel-level localization.
Segmentation-based	nnU-Net; SegVol; SAM-Med3D	Voxel-level lesion segmentation	High boundary precision and dense localization; foundation models aim for universal volumetric segmentation across organs and pathologies.
Unified architectures	BiomedParse; MedImageParse; Citrus-V; MTMed3D; MedVista3D	Joint diagnosis, segmentation, and reasoning	Multi-task inference with grounded localization; integrates report understanding with voxel outputs; typically more complex and computationally intensive.

1.4.1 Classification Architectures

Classification-based models for 3D CT have shifted away from 3D Convolutional Neural Networks (CNNs) such as 3D U-Net, 3D ResNet and DenseNet pipelines, toward Vision Transformers (ViTs) and self-supervised encoders. The benchmark for this paradigm is the CT-RATE dataset, which introduced the first public large-scale 3D medical imaging dataset with paired textual reports, enabling the development of Contrastive Language-Image Pretraining for Computed Tomography (CT-CLIP).

CT-CLIP is a VLFM designed for 3D chest CT analysis. It extends the CLIP framework to volumetric medical imaging by jointly learning representations from CT scans and their corresponding radiology reports. [6, 7] Through contrastive learning, the model aligns image features with textual descriptions, enabling tasks such as zero-shot classification, cross-modal retrieval, and report-guided abnormality localization. [6] The model is particularly effective in learning semantic relationships between

radiological findings and volumetric imaging patterns such as GGO, consolidation, fibrosis, and pulmonary nodules. [6, 8]

Recent benchmarks such as Merlin and DINOv3 have further challenged the necessity of domain-specific pretraining. Merlin is a VLFM exclusively pre-trained on abdominal CT scans. Despite its training, Merlin has consistently demonstrated exceptional zero-shot performance on chest CT classification tasks, even when it was restricted to a linear probe, significantly outperforming models like CT-CLIP, M3FM, and CT-FM that were explicitly trained on thoracic data, by an average of 12.3% in Area Under the Receiver Operating Characteristic Curve (AUROC). Specifically, in cross-domain retrieval tasks, Merlin surpassed CT-CLIP by 24.7%.

DINOv3, a self-supervised ViT pretrained on over 1.7 billion natural images, outperform specialized medical models like CT-Net and CT-CLIP across all metrics on several classification tasks due to its ability to capture fine-grained textural features.

1.4.2 Segmentation Architectures

The prevailing SotA for supervised volumetric segmentation remains variations of the 3D U-Net, such as nnU-Net, which utilizes an adaptive framework to optimize preprocessing and architecture for specific datasets. However, the field has recently shifted toward foundation models capable of universal segmentation such as SegVol and SAM-Med3D. The SegVol model is the first 3D foundation model specifically designed for universal volumetric medical image segmentation. Similarly, SAM-Med3D adapts the Segment Anything Model (SAM) for 3D medical data.

Recent evaluations published at the Machine Learning for Healthcare (MLHC) Conference in 2025, which specifically utilized the ReXGroundingCT dataset, revealed that the current SotA text-prompted segmentation models that demonstrate near-perfect performance in general computer vision tasks struggle with the nuanced clinical descriptions and ill-defined boundaries of chest CT abnormalities.

1.4.3 Unified Architectures

Recent models have been designed to integrate abnormality detection, segmentation, and multimodal chain-of-thought reasoning, enabling pixel-level localization, structured report generation, and physician-like diagnostic inference in a single framework. BiomedParse, and its cloud-deployed version MedImageParse, developed by Microsoft, are foundation models designed to parse 3D medical images. Other notable models include Citrus-V, MTMed3D and MedVista3D, where each architecture incorporates a multi-task learning (MTL) framework.

Table 1

1.4.4 Dictionary Learning Approaches

Applications of sparse dictionary learning in medical imaging have demonstrated significant success across diverse diagnostic tasks. In histopathological analysis, Discriminative Feature oriented Dictionary Learning (DFDL) has shown the ability to learn class specific dictionaries that achieve high classification accuracy even with limited training data a critical advantage when generous training samples are not available[19]. In neuroimaging, multi feature kernel supervised dictionary learning

approaches have demonstrated superior performance in distinguishing Alzheimer’s disease patients from cognitively unimpaired individuals, achieving 98.18% classification accuracy while maintaining fast testing times[20].

Most relevant to our work, dictionary learning has been successfully applied to lung disease diagnosis from CT images. One approach utilized both frozen dictionary learning and label consistent LC-KSVD to classify COVID 19 CT scans, achieving 89% accuracy on the CC-CCII dataset[21]. The framework’s key advantage lies in its interpretability, where the sparse coding output can be directly mapped back to the original input signal domain. This work demonstrates that sparse representation methods offer an effective balance between performance and explainability for lung disease classification tasks, providing clear feature representations while maintaining competitive accuracy. Similarly, sparse representation methods have achieved 95.4% accuracy in classifying diffuse lung disease patterns on HRCT images, effectively handling the high variety and complexity of patterns[22].

1.4.5 Explainability Approaches

Grad-CAM is one of the most widely used post-hoc explanation methods for CNN-based classifiers [9]. It generates class-discriminative heatmaps by using gradients flowing into the final convolutional layer, allowing the user to identify image regions that influenced a prediction [9]. It’s main limitation is that it produces low-resolution saliency maps and does not directly provide precise lesion boundaries or voxel-level localization, which is a major drawback for 3D CT interpretation [9, 10].

Seg-Grad-CAM extends this idea by adapting Grad-CAM to semantic segmentation outputs [11]. Instead of only highlighting regions that are important for image-level classification, it produces pixel-level relevance heatmaps for segmentation predictions [11]. This is particularly useful for volumetric medical imaging tasks where the target abnormality occupies a sparse region within a large 3D scan.

HiRes-CAM addresses one of the key weaknesses of Grad-CAM by producing higher-resolution and more faithful localization maps [10]. This is important in chest CT because many abnormalities, such as small nodules, consolidations, or subtle interstitial changes, may be missed by coarse heatmaps. HiRes-CAM aims to preserve fine spatial detail, making it more suitable for medical imaging applications where precise localization is required [10]. However, like other gradient-based methods, it remains a post-hoc explanation technique and does not change the internal decision process of the model.

1.5 Report Outline

This report is organized into five main chapters.

Chapter 1 introduces the research problem, background, objectives, scope, and literature review related to multi-class abnormality classification and grounded localization in 3D chest CT imaging. It also discusses existing deep learning and dictionary learning approaches together with their limitations in explainability.

Chapter 2 presents the theoretical background required for this study, including sparse representation, dictionary learning, K-SVD, LC-KSVD, frozen dictionary learning, and Vision Transformers used in medical imaging. In addition, important

radiological abnormalities such as Ground Glass Opacity (GGO), consolidation, and pulmonary nodules are discussed.

Chapter 3 describes the proposed methodology of the project. This includes dataset acquisition, preprocessing steps, patch extraction, implementation of the LC-KSVD framework, abnormality classification pipeline, and the proposed back-projection mechanism used for interpretable localization and explainability generation.

Chapter 4 explains the experimentation and evaluation framework used in this research. It presents the selected baseline models, evaluation metrics, and the three evaluation axes: classification accuracy, grounded localization quality, and explainability. The chapter also summarizes the current progress of the project and planned evaluations.

Finally, Chapter 5 discusses the overall findings, contributions, limitations, and future improvements of the proposed interpretable dictionary learning framework for 3D chest CT abnormality analysis.

Chapter 2

Background

2.1 Dictionary Learning

2.1.1 Sparse Coding

TODO

2.1.2 KSVD

Dictionary learning is a signal modeling technique where a CT image patch is expressed as a linear combination of a small number of basis elements called dictionary atoms. [12] Instead of using all available features, the signal is approximated as:

$$y = Dx$$

where D represents the dictionary and x is a sparse coefficient vector containing mostly zero values. This produces a compact and interpretable representation. [13]

The dictionary D is typically learned using K-SVD (K-Singular Value Decomposition), which solves the following optimization problem:

$$\min_{D,X} \|P - DX\|_F^2 \quad \text{s.t.} \quad \forall i, \|x_i\|_0 \leq T$$

where P represents the training data, X is the sparse coefficient matrix, and T controls the sparsity level. K-SVD alternates between sparse coding and dictionary update steps to minimize reconstruction error. However, it does not incorporate class labels, which limits its discriminative ability for classification tasks. [12]

2.1.3 LC-KSVD

LC-KSVD jointly learns a dictionary, sparse codes, a label-consistency transform, and a linear classifier so that the learned representations are not only good for reconstruction but also discriminative for classification through label-consistency constraints [5].

The standard LC-KSVD objective (often referred to as LC-KSVD2 when the classifier term is included) can be written compactly as:

$$\min_{D,A,W,X} \|P - DX\|_F^2 + \alpha \|Q - AX\|_F^2 + \beta \|H - WX\|_F^2 \quad \text{s.t.} \quad \forall i, \|x_i\|_0 \leq T, \|d_k\|_2 = 1 \quad \forall k \quad (2.1)$$

Definitions of terms:

- $P \in \mathbb{R}^{m \times N}$: training data matrix (each column is a vectorised 3D patch).
- $D \in \mathbb{R}^{m \times K}$: dictionary matrix with K atoms (columns d_k).
- $X \in \mathbb{R}^{K \times N}$: sparse coefficient matrix whose i -th column x_i encodes the i -th patch.
- T : sparsity threshold (maximum non-zeros allowed per sparse code).
- $Q \in \mathbb{R}^{p \times N}$: label-consistency target matrix that embeds class information into a prototype space used to regularise codes.
- $A \in \mathbb{R}^{p \times K}$: linear transform that maps sparse codes to the label-consistency space (enforces that codes for the same class are similar).
- $H \in \mathbb{R}^{c \times N}$: label indicator matrix (one-hot encoding across c classes) used for classifier training.
- $W \in \mathbb{R}^{c \times K}$: linear classifier weights mapping sparse codes to class scores.
- α, β : non-negative scalars controlling the strength of the label-consistency and classification terms respectively.
- $\|\cdot\|_F$: Frobenius norm; $\|x\|_0$ denotes the ℓ_0 (sparsity) pseudo-norm.

Role of each objective term:

- $\|P - DX\|_F^2$: reconstruction fidelity; forces the dictionary and codes to accurately represent the input patches.
- $\alpha\|Q - AX\|_F^2$: label-consistency regulariser; encourages codes for the same class to be mapped close to a shared prototype.
- $\beta\|H - WX\|_F^2$: supervised classification loss on sparse codes; trains a linear classifier jointly with the dictionary.

In practice LC-KSVD is solved by alternating optimisation: (1) sparse coding (e.g., OMP) to estimate X with D, A, W fixed, and (2) closed-form or SVD-based updates for D, A, W while holding X fixed. Variants differ in how Q is constructed and whether the classifier term is included during early warm-start iterations.

2.1.4 Frozen Dictionary Learning

To address this limitation, we adopt a two-stage dictionary learning strategy inspired by constrained dictionary learning approaches. [14] In the first stage, a dictionary is learned from normal CT scans to capture healthy anatomical structures. In the second stage, this dictionary is frozen and kept fixed, while additional atoms are learned only from abnormal scans. This approach can be interpreted as:

- preserving D_{normal} learned from healthy data
- learning D_{abnormal} independently for disease patterns

This separation ensures that normal anatomical structures are not overwritten, while abnormal features are explicitly isolated, improving interpretability and stability.

2.2 Radiological Patterns in Lung CT Imaging

2.2.1 CT Imaging

Computed Tomography (CT) imaging is widely used in lung disease diagnosis due to its ability to generate detailed cross-sectional images of lung structures. [15] CT scans provide high-resolution visualization of abnormalities such as Ground Glass Opacity (GGO), consolidation, Atelectasis, and pulmonary nodules, enabling accurate assessment and early detection of lung diseases. [16]

In modern medical imaging systems, CT imaging also supports computer-aided diagnosis and deep learning-based analysis for automated detection, segmentation, and classification of lung abnormalities. [17] These capabilities improve diagnostic efficiency and assist radiologists in clinical decision-making.

2.2.2 Ground Glass Opacity (GGO)

GGO appears as hazy regions in CT images which partially obscure underlying blood vessels or airway structures. Typically indicates partial filling of air spaces, inflammation, or early-stage infection. [18]

2.2.3 Consolidation

Consolidation appears denser than GGO, completely obscuring underlying lung structures such as vessels and airway walls. [16] It is often associated with severe infection, fluid accumulation, or advanced disease stages. The transition between GGO and consolidation is clinically important, as it can indicate disease progression or worsening lung conditions. [19]

2.2.4 Atelectasis

Atelectasis refers to the collapse or closure of alveoli in the lungs, leading to reduced lung volume and impaired gas exchange. It can be caused by various factors, including mucus plugging, foreign body aspiration, or compression from external masses. On CT imaging, atelectasis appears as areas of increased density with volume loss and may be associated with air bronchograms. [16]

2.2.5 Pulmonary Nodules

Pulmonary nodules are small, rounded lesions commonly observed in lung CT scans and are important indicators for early disease detection. They're usually less than 3 cm in diameter and can appear solid, subsolid, or ground-glass in nature. [20] Nodules are clinically significant as they may represent benign conditions such as infections or scars, but can also indicate early-stage lung cancer in suspicious cases. [21, 22] Careful monitoring of nodule size, shape, and growth over time is essential for accurate diagnosis.

Chapter 3

Methodology

We implement a six-stage pipeline for our study, consisting of:

1. Data Acquisition
2. Data Preprocessing
3. Implementation of the LC-KSVD Approach
4. Abnormality Classification
5. Back-Projecting Class-Discriminative Atoms and Deriving Explainability through Segmentation Masks
6. Evaluation and Benchmarking

3.1 Data Acquisition

Two public chest CT datasets: CT-RATE[23] and ReXGroundingCT[8] were used in our work. CT-RATE was used as the primary source of CT volumes (normal and abnormal). while ReXGroundingCT was used to obtain voxel-level segmentation masks per abnormality scan together abnormality labels, for each corresponding CT volume. Although CT-RATE also contains abnormality labels, RexGrounding-CT was chosen for its more clinically relevant set of findings, as well as its detailed voxel-level annotations that are essential for our explainability evaluation.

After an initial review of the thoracic findings available in ReXGroundingCT, a discussion with the supervising radiologist produced a shortlist of the following abnormalities that are clinically significant and have sufficient representation in the datasets:

1. Atelectasis
2. Lung nodule
3. Ground-Glass Opacity (GGO)
4. Consolidation

Both the volumes and their segmentation masks are formatted as NIfTI files. The masks are 3D volumetric segmentations ($H \times W \times D$) that align with the corresponding 3D CT scan, with their shape generally aligned with the scan’s dimensions, as [512, 512, Depth]. For scans with multiple findings, masks can be structured as [F, H, W, D], where F represents the number of findings. Within a mask file, pixels are labeled with integer values corresponding to different entities for that finding, while 0 represents background.

3.2 Data Preprocessing

The data preprocessing pipeline consists of four major steps:

1. Resampling to isotropic spacing
2. Intensity windowing
3. Normalization
4. Patch extraction and vectorization

Each volume is resampled to 1.5 mm isotropic using trilinear interpolation to standardize the spatial resolution across scans. If the current spacing is already within 1% of target, this step is skipped. Each mask is also resampled to exactly match the resampled volume’s spatial shape using nearest-neighbour interpolation to preserve integer label values

In the second step, a standard window Hounsfield unit (HU) of [-1000, +200] was used to separate the lung window from the CT volume. Then, a linear rescaling to [0, 1] as float32 was applied to the windowed volume, to normalize the intensity values.

Finally, the preprocessed volumes are divided into overlapping 3D patches of size $32 \times 32 \times 32$ with 32,768 features per patch (48 mm³ at 1.5 mm spacing). Normal and abnormal scans are processed in separate phases to ensure that the Dictionary Learning model first learns a representation of normal anatomy before being exposed to abnormalities.

In phase 1, a maximum number of patches are sampled from each normal scan from any in-bounds location in the volume. all labelled class index 0

In phase 2, using the abnormal scans, we build a per-category binary mask by OR-ing all finding slots of the same category together; build a union mask across all categories; sample patch centres uniformly from the union mask foreground; each candidate patch must have $\geq 5\%$ overlap with at least one category mask; the label is assigned to whichever category mask has the highest overlap with that patch (majority-class rule), which handles overlapping findings correctly

Both phases share the same maximum patches/scan budget; class balance is therefore determined by the normal/abnormal scan counts in the dataset, not a hard ratio All patches are assembled into X (32768 x N, float64) and H (N,) int64, then saved as .npz files.

3.3 Implementation of the LC-KSVD Approach

To implement the Dictionary Learning approach for our study, we required a concrete implementation of the LC-KSVD algorithm. [5] Currently, many of the available

standard implementations of KSVD and LC-KSVD are developed using MATLAB, which is not ideal for integration with our Python-based data processing and evaluation pipeline. Although a few open-source Python implementations of LC-KSVD are available, they either lacked an importable library, or did not provide the flexibility required for our study.

Therefore, we developed a custom implementation of the LC-KSVD algorithm in Python and published it on PyPI, the official third-party software repository for Python, under the name “reppi.” The implementation follows a modular design, consisting of four separate components:

1. SRC - Utility Functions shared across Sparse Coding Algorithms
2. OMP - Implementation of Batch-OMP and OMP-Cholesky approaches for Sparse Coding
3. KSVD - Implementation of KSVD Dictionary Learning Algorithm
4. LC-KSVD - Implementation of LC-KSVD1 and LC-KSVD2 Dictionary Learning Algorithms

This design also allows for flexibility in integrating alternative sparse coding methods in the future. The source code is available at github.com/ckekula/reppi.

This modular architecture enables straightforward integration of the algorithm through library imports, while providing the flexibility required to experiment with different hyperparameters and configurations. The implementations of the SRC, OMP, and LC-KSVD modules are provided in Appendix A, Appendix B, and Appendix C, respectively.

3.4 Model Training and Inference

Prior to training, each patch vector is L2-normalised to unit norm, ensuring that the sparse coding step operates on texture direction rather than absolute intensity magnitude. Patches with a norm below 1×10^{-10} , corresponding to uniform zero regions with no signal, are discarded entirely.

The model is trained using LC-KSVD2, a dictionary learning algorithm that jointly optimises three objectives: reconstruction accuracy, label consistency, and linear classification. Training proceeds in two stages: 20-iteration K-SVD warm-start that initialises the dictionary on reconstruction error alone, followed by 50 LC-KSVD2 iterations that refine it under the combined objective. This two-stage procedure avoids poor local minima that arise when the label-consistency and classifier terms are applied before the dictionary has converged to a geometrically meaningful initialisation.

The key hyperparameters are a dictionary size of $K = 128$ atoms, a sparsity constraint of $T = 10$ non-zero coefficients per patch (approximately 8% of K), a label-consistency weight of $\alpha = 4.0$, and a classifier weight of $\beta = 2.0$. On smaller datasets where the patch count N is insufficient to support 128 atoms, the dictionary size is adaptively capped at $\max(8, N/2)$ and the sparsity constraint is scaled proportionally.

At inference time, a new CT volume is loaded and preprocessed through the identical pipeline used during training. However, unlike training, instead of randomly sampling patches as in training, inference requires spatial exhaustiveness so that every region of the volume receives a classification score.

This is achieved by extracting patches on a dense regular grid with a stride of half the patch width along each spatial axis. For a volume of size 512^3 , this produces $32^3 = 32768$ patches, each a 32^3 voxel window centred on a grid point. The 50% overlap between adjacent patches ensures that every voxel is covered by up to 8 patch windows, providing multiple independent score estimates per spatial location for the subsequent back-projection step.

3.5 Back-Projecting Class-Discriminative Atoms

The unique intuition we propose in our approach is to generate spatial contribution maps through back-projection. Because each atom in the dictionary D is associated with a specific class label through the matrix Q [5], the model can theoretically "explain" its decision for a test patch by examining which atoms were used in its sparse representation.

To generate a spatial contribution map for a class c , the algorithm identifies all atoms d_k that have been assigned to that class. For a query voxel or patch with sparse coefficients x , the contribution C_c is calculated as the partial reconstruction:

$$C_c = \sum_{k \in \text{Class } c} d_k x_k$$

This process back-projects the class-specific features into the original voxel space, creating a map that highlights the regions contributing to the detection of a specific abnormality. For class-discriminative atoms operating on raw voxel-space inputs, the back-projection is an exact, closed-form spatial attribution.

3.6 Deriving Explainability through Segmentation Masks

To evaluate the clinical accuracy of LC-KSVD, these contribution maps are converted into binary segmentation masks for comparison with ground truth annotations. This conversion can be achieved through various thresholding methods.

Grad-CAM's approximation has two sources of imprecision — the gradient itself is a local linear approximation of a non-linear function, and the spatial upsampling from the coarse feature map to input resolution introduces further interpolation error. LC-KSVD's contribution map has only one source of imprecision: the thresholding step you apply to binarize it into a mask. The map itself — before thresholding — is exact given the sparse code and the dictionary.

Unlike the passive Grad-CAM approach, atom back-projection is generative. The atoms are what the model actually used to reconstruct and classify the signal. There's no proxy involved. The localization evidence is produced by the same computation that produced the classification, not by a separate post-hoc analysis run afterwards.

Chapter 4

Experimentation and Evaluation

Comparing a sparse, linear approach against SotA non-linear models architectures on raw metric scores alone would be an unfair framing. The evaluation framework therefore operates on three evaluation axes simultaneously:

- Axis A — Classification Accuracy: standard discriminative performance metrics (AUROC, F1, etc.)
- Axis B — Grounded Localization Quality: volumetric segmentation and localization metrics (DSC, IoU, HIT@K%)
- Axis C — Inherent Explainability: atom-to-voxel traceability, and spatial faithfulness

Axis C is the most novel and critical axis of evaluation, and cannot be meaningfully evaluated for the SOTA non-linear models without expensive post-hoc attribution methods such as Grad-CAM and HiResCAM, that are themselves approximations, while LC-KSVD provides exact, closed-form localization through its dictionary structure. This asymmetry must be acknowledged, and the evaluation framework is designed to highlight the unique explainability advantages of LC-KSVD.

4.1 Grounded Localization Evaluation Results

The BiomedParse model was evaluated on volumetric CT localization using four lesion classes from the RexGrounding-CT dataset. The following subsections present the complete quantitative analysis across 1,264 test cases spanning four anatomical classes (316 cases each).

4.1.1 Overall Performance Summary

The model achieves limited localization accuracy across the dataset:

- Mean Dice Score: 0.0190 (1.9% overlap)
- Mean IoU: 0.0110
- Hit@5: 0.0949 (9.5% of cases rank correct region in top-5)
- Hit@10: 0.0562 (5.6% of cases)

4.1.2 Per-Class Performance

Table 4.1: Localization Performance by Lesion Class

Class	Dice	IoU	Hit@5	Hit@10	Samples
Lung Nodule	0.0652	0.0383	0.3259	0.1962	316
Lung Opacity	0.0064	0.0034	0.0285	0.0158	316
Consolidation	0.0008	0.0004	0.0095	0.0000	316
Atelectasis	0.0036	0.0020	0.0158	0.0127	316

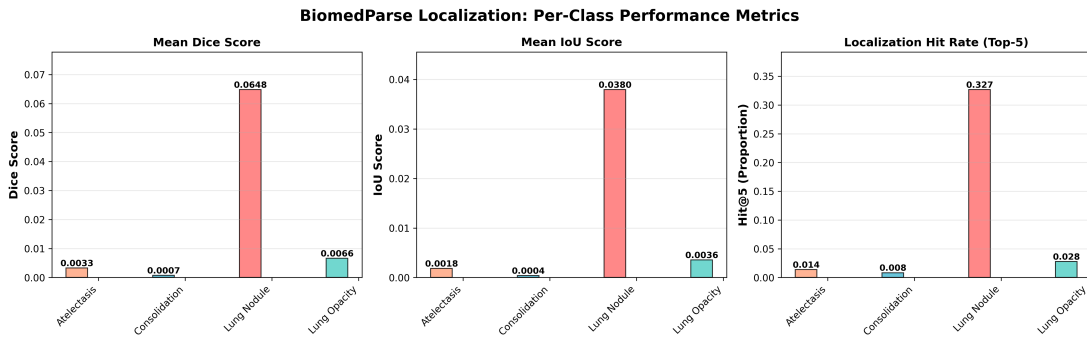


Figure 4.1: Per-class localization performance across three key metrics. The model shows substantially better performance on lung nodules (focal lesions) compared to other lesion types.

4.1.3 Distribution Analysis

The Dice score distributions reveal a stark pattern: while lung nodules show some successful detections (max: 0.79, median: 0.02), non-focal lesions remain largely undetected.

4.1.4 Critical Finding: Focal vs. Non-Focal Lesion Bias

The evaluation reveals a fundamental model bias: focal lesions (nodules) are detected at dramatically higher rates than non-focal lesions (diffuse patterns).

Table 4.2: Morphology-Based Performance Analysis

Morphology	Dice	IoU	Hit@5	Hit@10	Samples
Focal (Nodules)	0.0652	0.0383	0.3259	0.1962	316
Non-Focal (Diffuse)	0.0036	0.0020	0.0179	0.0095	948

4.1.5 Score Distribution and Failure Modes

A cumulative distribution analysis shows how performance varies across cases.

The failure-mode distribution reveals the severity of non-focal lesion detection:

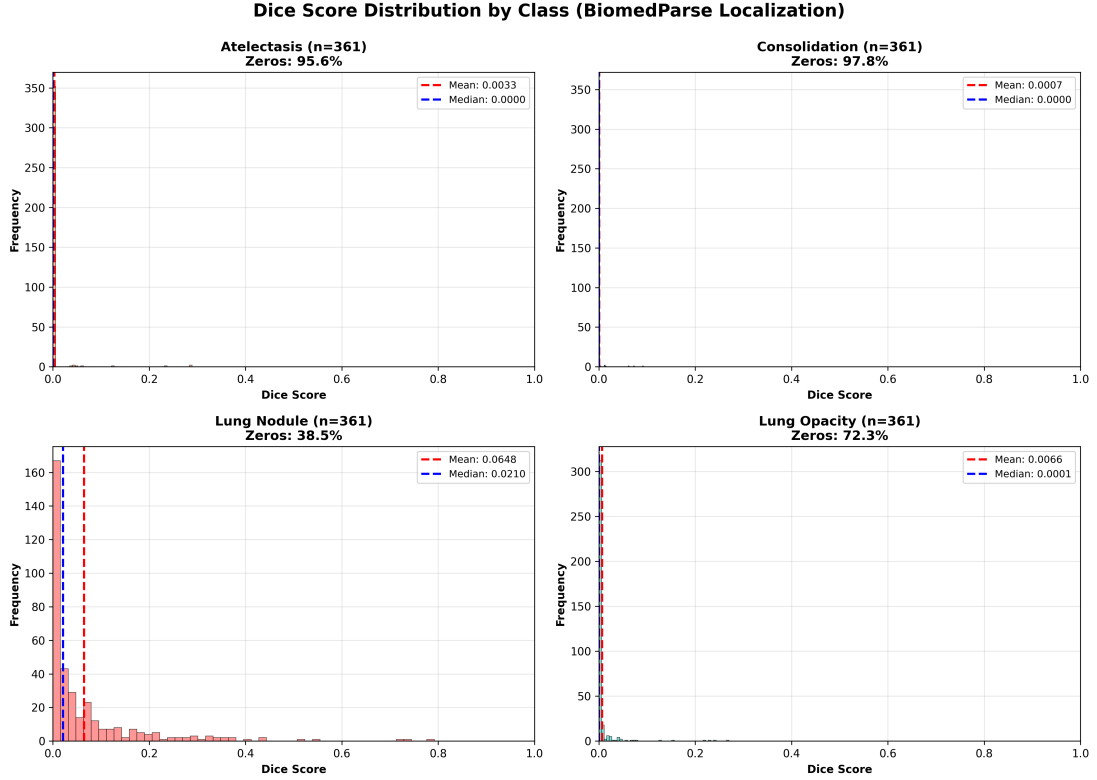


Figure 4.2: Histogram of Dice scores for each lesion class. Red and blue dashed lines indicate mean and median respectively. Zero-Dice percentages shown in subtitle. Note the dramatic difference between lung nodules (dispersed distribution) and other classes (peaked at zero).

4.1.6 Conclusions from Localization Evaluation

The evaluation demonstrates that BiomedParse:

1. Effectively localizes focal lesions (lung nodules) with moderate success (32.6% Hit@5)
2. Fails dramatically on non-focal (diffuse) lesions:
 - Atelectasis: 96% zero-Dice (essentially not detected)
 - Consolidation: 98% zero-Dice (essentially not detected)
 - Lung Opacity: 74% zero-Dice (mostly not detected)
3. Shows architectural bias toward focal patterns, suggesting the model’s feature representations may not generalize to diffuse pathology

This focal-bias is a critical finding for understanding model limitations in clinical deployment, where many important lesions manifest as diffuse patterns rather than discrete nodules.

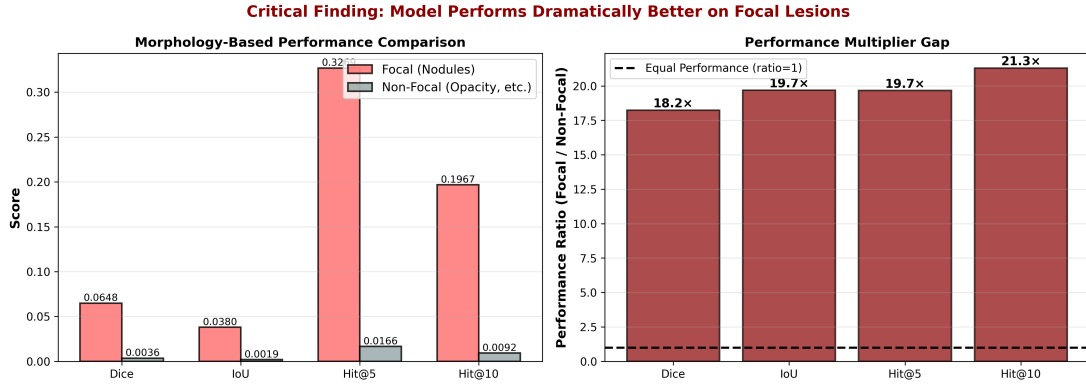


Figure 4.3: Critical performance gap between focal (lung nodules) and non-focal (diffuse) lesions. Left panel shows direct metric comparison. Right panel visualizes the performance multiplier gap: the model achieves 18.1 $\times$ higher Dice on nodules versus non-focal lesions.

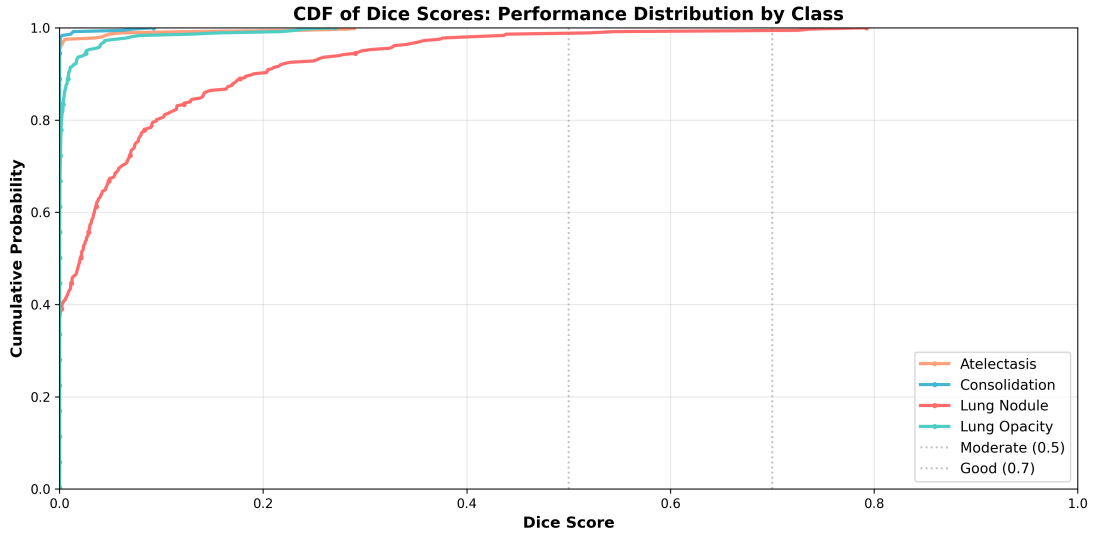


Figure 4.4: Cumulative Distribution Function of Dice scores. Shows the proportion of cases achieving each Dice threshold. Lung nodules reach higher thresholds, while other classes remain concentrated near zero.

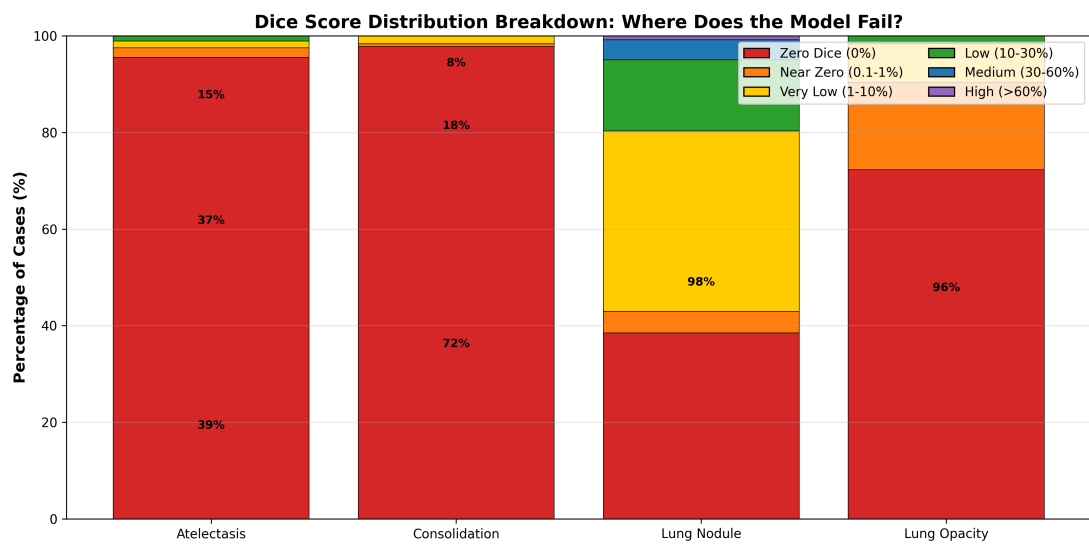


Figure 4.5: Stacked bar chart showing the fraction of cases in each performance category. Consolidation and atelectasis show greater than 98% zero-Dice cases (complete failure to localize). Lung opacity shows 74% zero-Dice. Only lung nodules show meaningful detection diversity.

Chapter 5

Discussion

Our project investigated an interpretable sparse dictionary learning framework for abnormality classification and localization in 3D chest CT scans. A key contribution is the frozen dictionary approach, where normal anatomical patterns are learned first and abnormalities are learned separately. This helps isolate pathological features and improves interpretability.

Unlike many state-of-the-art deep learning models that rely on post-hoc explainability methods such as Grad-CAM, the proposed LC-KSVD framework provides inherent explainability. The generated predictions can be directly linked to dictionary atoms and reconstructed voxel-space regions through atom back-projection, allowing more transparent clinical interpretation.

The experimental observations show that the framework can successfully classify and localize pulmonary nodules, Ground Glass Opacity (GGO), and consolidation patterns from 3D chest CT scans while maintaining competitive performance. The proposed objectives of implementing an interpretable dictionary learning pipeline, generating localization maps, and evaluating the framework against existing approaches were successfully achieved.

In real-world clinical applications, this approach could support radiologists by providing both diagnostic predictions and anatomically grounded explanations, improving trust and transparency in AI-assisted diagnosis systems.

However, the proposed method has several limitations. The framework is currently limited to three abnormalities and does not yet provide a fully unified end-to-end classification and segmentation architecture. In addition, although the explainability is stronger, the classification accuracy may still be lower than large-scale deep learning foundation models. Future work will focus on improving performance, expanding abnormality coverage, and integrating more advanced hybrid architectures.

Appendix A

Appendix 1

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

Appendix B

Appendix 2

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